D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Adverse events in CAR-T cell therapy

August 2022 | DM 0822 | Q4 (Short note / Write notes) | **7 marks** | Target ~11 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

1. Introduction & exam framing

This answer addresses: **Adverse events in CAR-T cell therapy**

For a **7-mark** university answer, structure the response as: definition/epidemiology → biology/hallmarks → diagnosis & biomarkers → staging/risk → management algorithm with doses → key clinical trials with endpoints → toxicities/supportive care → **NCCN 2026 updates** → Indian/NCG context → summary table. Draw at least one flowchart and one mechanism diagram.

Marking examiners reward: accurate mechanisms, named trials with phase and primary endpoint, practical doses, Category evidence (NCCN/ESMO), and a clear final recommended pathway.

- Start with a one-line definition + why the topic is ‘recent advances’ relevant.
- Always state predictive vs prognostic biomarkers.
- Quote Category 1 preferred regimens from NCCN 2026/2027 when applicable.
- End with 5 high-yield exam pearls.

2. Pathobiology, hallmarks & mechanism

Cancer biology for this topic should be linked to Hanahan–Weinberg hallmarks (sustained proliferation, evading growth suppressors, resisting cell death, enabling replicative immortality, induction of angiogenesis, activating invasion/metastasis, avoiding immune destruction, deregulating cellular energetics, genome instability, tumor-promoting inflammation).

Where a driver oncogene/fusion/epigenetic lesion is known, explain the pathway from ligand/receptor → intracellular kinase cascade → transcription → phenotype, and mark the drug target with an arrow in your diagram.

Adoptive cellular therapy redirects immunity: CAR-T grafts an antigen binder + signaling domains (CD3ζ ± 4-1BB/CD28) onto T cells; BiTEs are off-the-shelf engagers linking CD3 to tumor antigen (CD19/BCMA/GPRC5D). Toxicities CRS/ICANS are on-target immune activation.

Driver / target	State oncogene, fusion, receptor, or immune checkpoint involved
Pathway	MAPK / PI3K-AKT-mTOR / JAK-STAT / DDR / BCR signaling / angiogenesis
Hallmark effect	Proliferation, survival, angiogenesis, immune escape, or DNA repair addiction
Drug class	mAb / TKI / ADC / BiTE / CAR-T / CPI / PARPi / IMiD / BH3 mimetic
Resistance	Secondary mutation, bypass track, phenotypic switch, antigen loss, TME suppression

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

• ASCO/CAP: IHC 0/1+/2+/3+; ISH if 2+; HER2-low = IHC 1+ or 2+/ISH–

- Draw the pathway once; label drug targets — examiners love this.
- Always mention one primary resistance mechanism.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

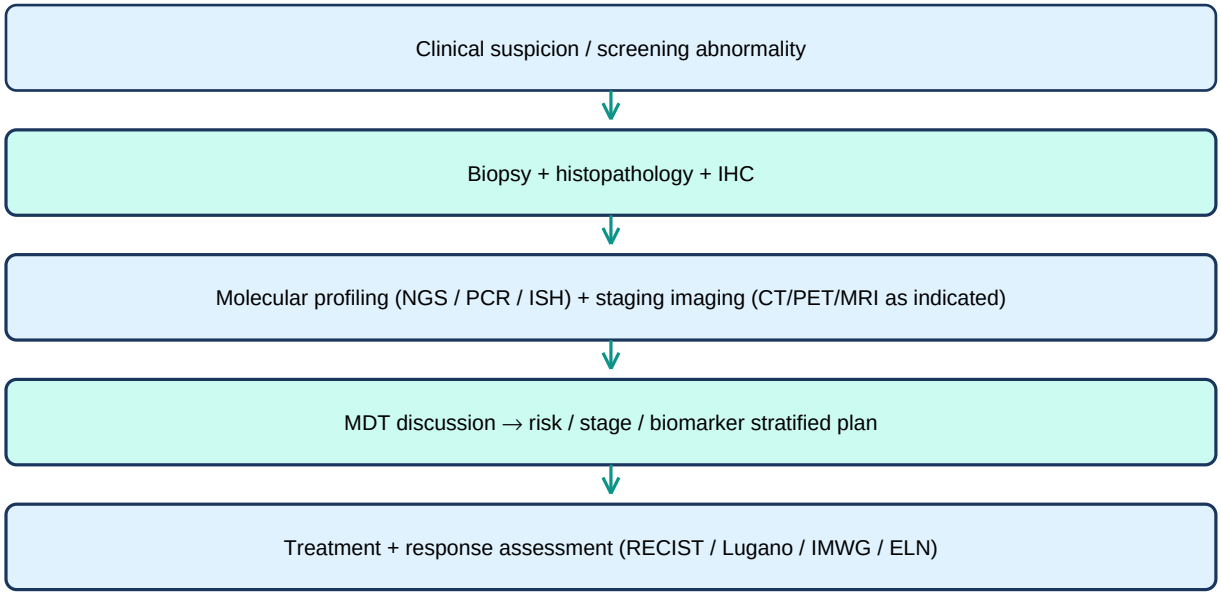

Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.
- Indications: R/R LBCL, B-ALL, mantle, follicular (product-specific), myeloma BCMA CAR-T.

- Pathway: apheresis → manufacturing → lymphodepletion (Flu/Cy) → infusion → CRS/ICANS watch.
- Bridging therapy may be needed; manufacturing failure contingency.

Management flowchart (reproduce in exam)

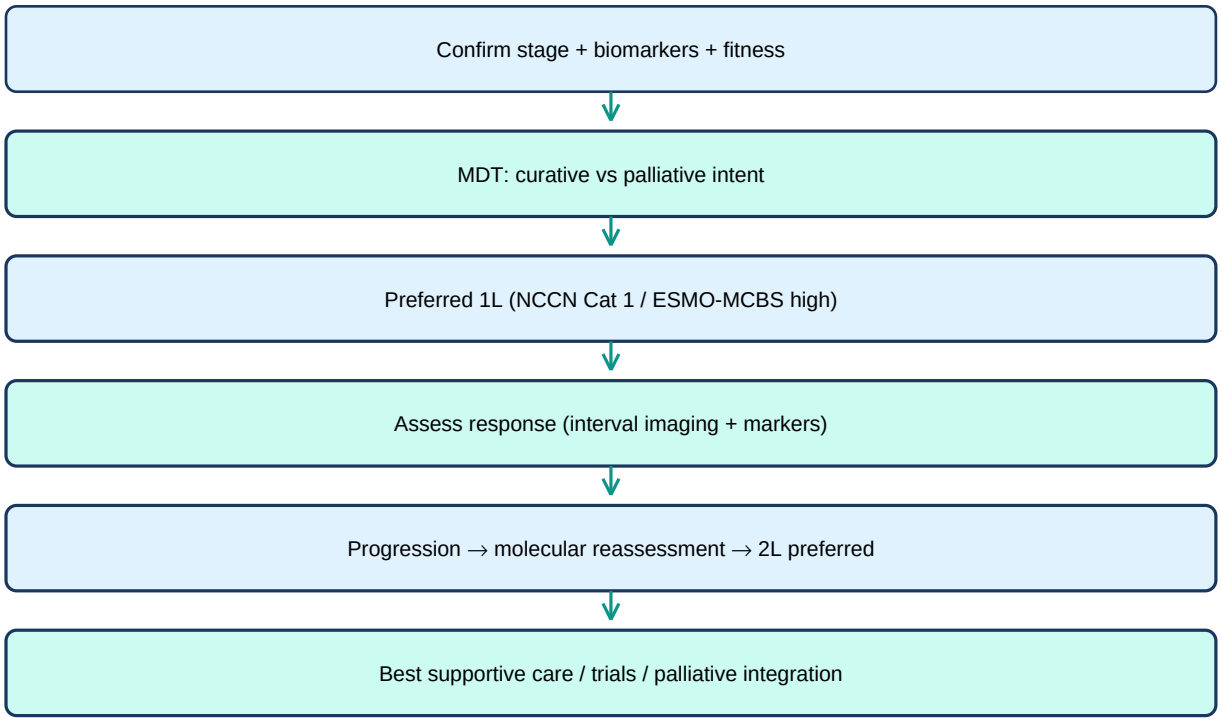

Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC
ELIANA / ZUMA-3	ALL CAR-T
CARTITUDE / KarMMa	BCMA CAR-T myeloma

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.
- CRS: tocilizumab ± dexamethasone; ICU for grade ≥3.
- ICANS: ICE score q shift; dexamethasone; seizure prophylaxis in high-risk protocols; neuroimaging/EEG PRN.
- Prolonged cytopenias, hypogammaglobulinemia (IVIG), infections — PJP/VZV prophylaxis per center.

Toxicity management loop

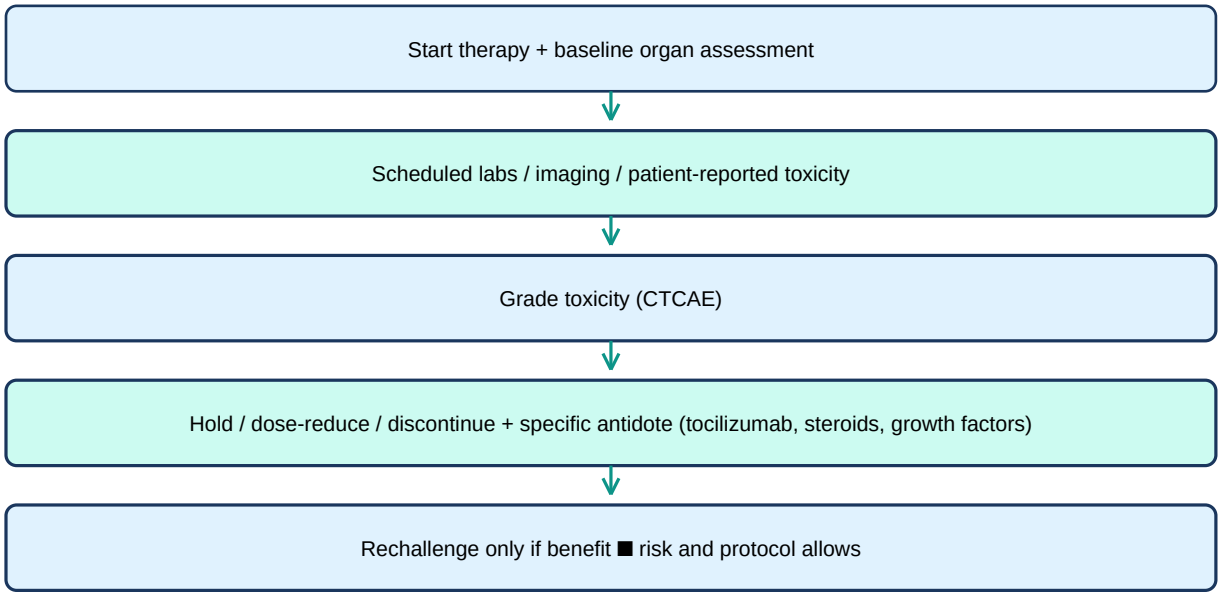

Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Multiple Myeloma v2.2027; B-Cell Lymphomas v4.2026
- Isa-VRd Cat 1 preferred induction for transplant-eligible newly diagnosed MM; Dara-VRd Cat 1 preferred for transplant-deferred/<80 y pathways.
- Maintenance daratumumab useful in certain circumstances post-transplant.
- Early relapse: teclistamab + daratumumab Cat 1 preferred for PI/IMiD-refractory after ≥1 prior line including len + PI (MajesTEC-3: 36-mo PFS 83.4% vs 29.7%).
- Belantamab mafodotin + Vd Cat 1 other recommended after ≥2 prior (PI + IMiD) — DREAMM-7.

• Later lines: linvoseltamab preferred after ≥4 prior lines including anti-CD38, PI, IMiD; talquetamab may bridge to BCMA CAR-T.

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

• Earlier-line CPI expansion: ATOMIC stage III dMMR colon; neoadjuvant CPI for dMMR rectal; neoadjuvant melanoma; EV+P bladder; FLOT+durvalumab gastric.

• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

EXAM PEARLS / HALLMARKS

• NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).

• Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

• MDT documentation improves outcomes and exam completeness.

• Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.

• Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

• Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.

• Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.

• Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.

• Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

• Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)

• Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 7-mark answer with a crisp numbered summary the examiner can tick.

• Definition + epidemiology one-liner

• Key biology / hallmark diagram

• Essential diagnostics & biomarkers

• Risk/stage stratification

- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

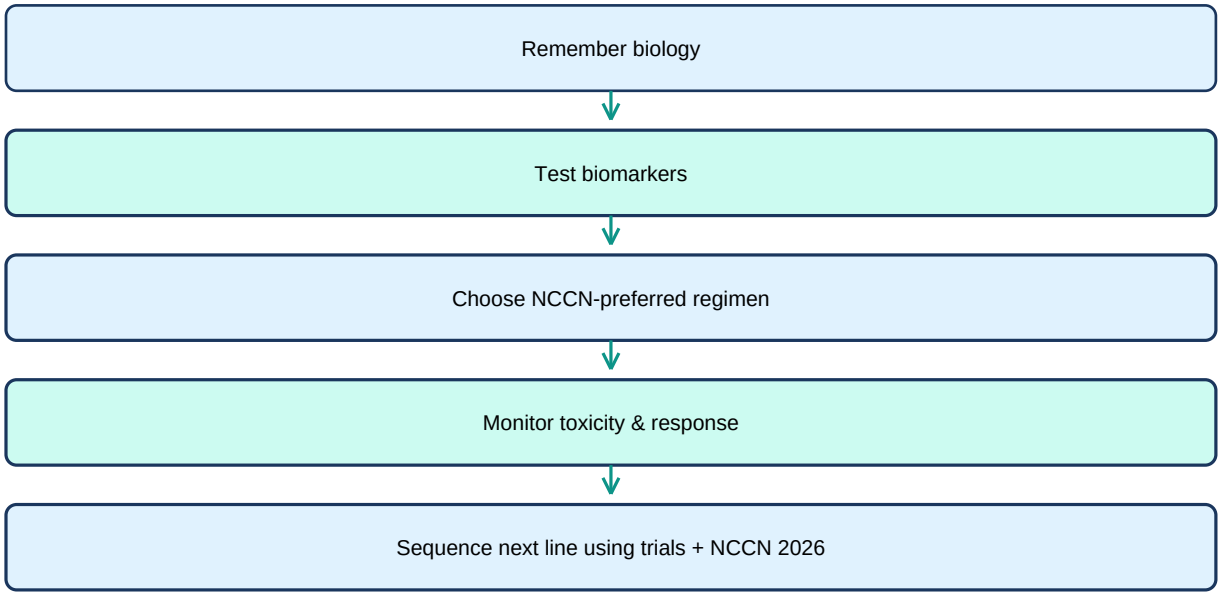

Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.
- ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

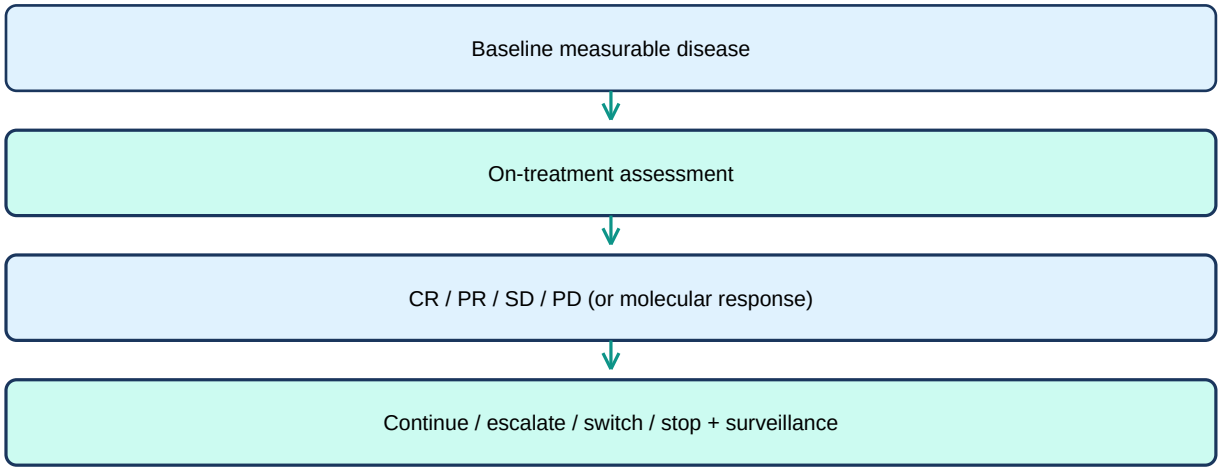

Figure: stepwise exam algorithm

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

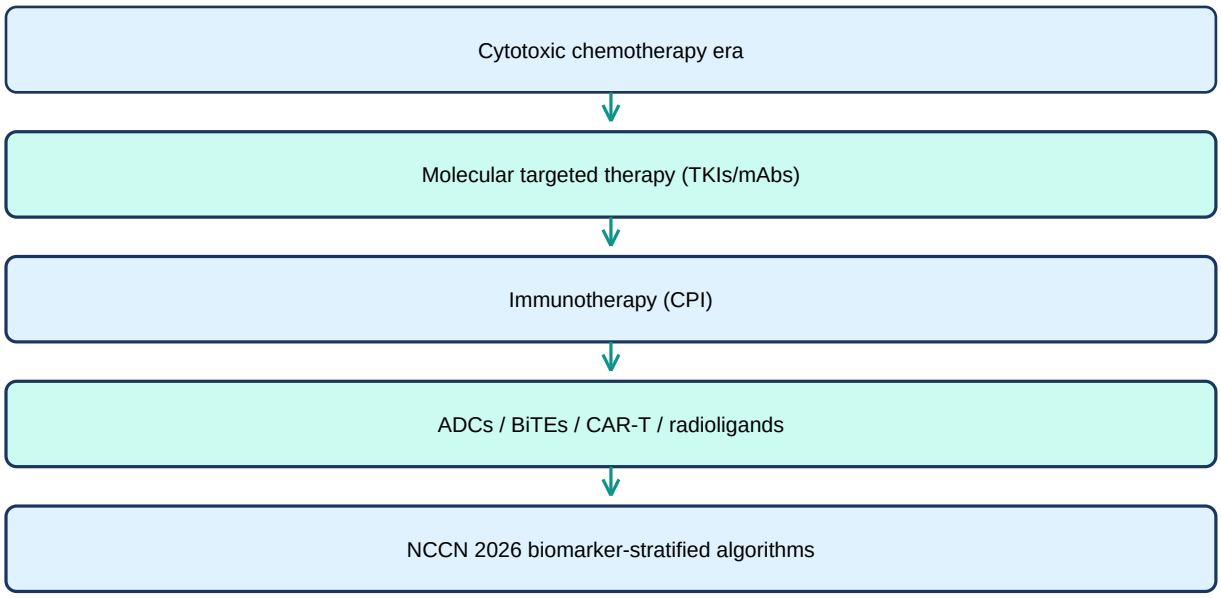

Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

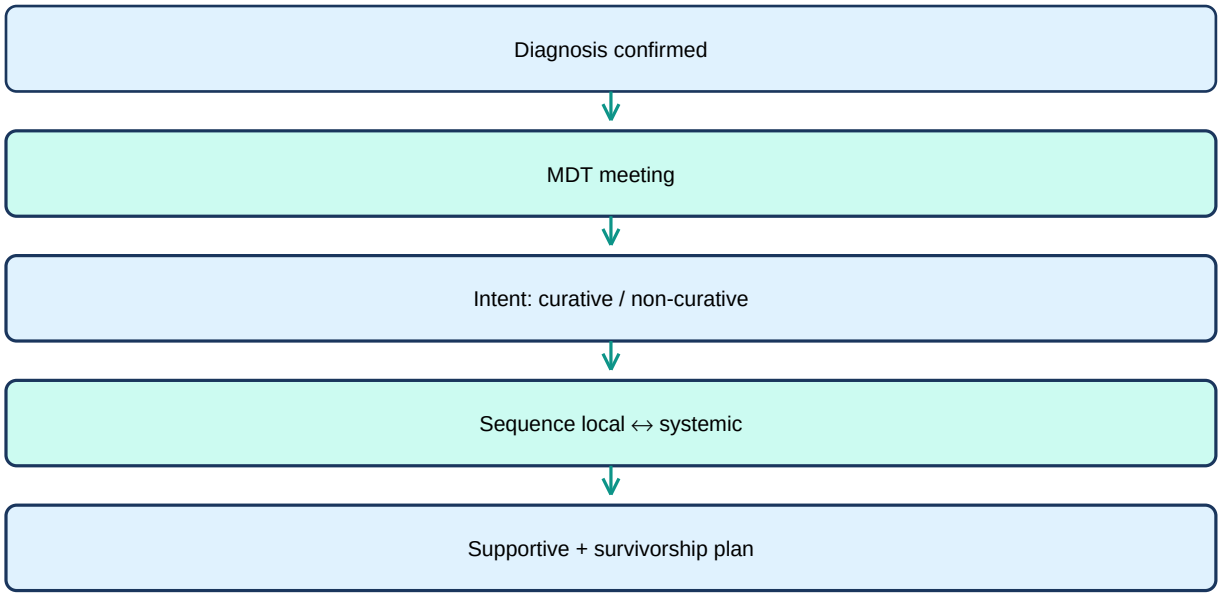

Figure: stepwise exam algorithm

E. Cellular therapy & engagers deep notes

CAR structure: scFv + hinge/transmembrane + costim (4-1BB or CD28) + CD3ζ. Manufacturing autologous vs emerging allogeneic/CAR-NK.

BiTEs continuous infusion (blinatumomab) vs newer half-life extended engagers; step-up dosing to mitigate CRS.

ASTCT CRS/ICANS grading mandatory knowledge. Tocilizumab for CRS; steroids for ICANS; anakinra in refractory protocols.

NCCN pathways place CAR-T earlier in LBCL (ZUMA-7) and after BTKi/BCL2i in CLL (liso-cel).

Clinical algorithm

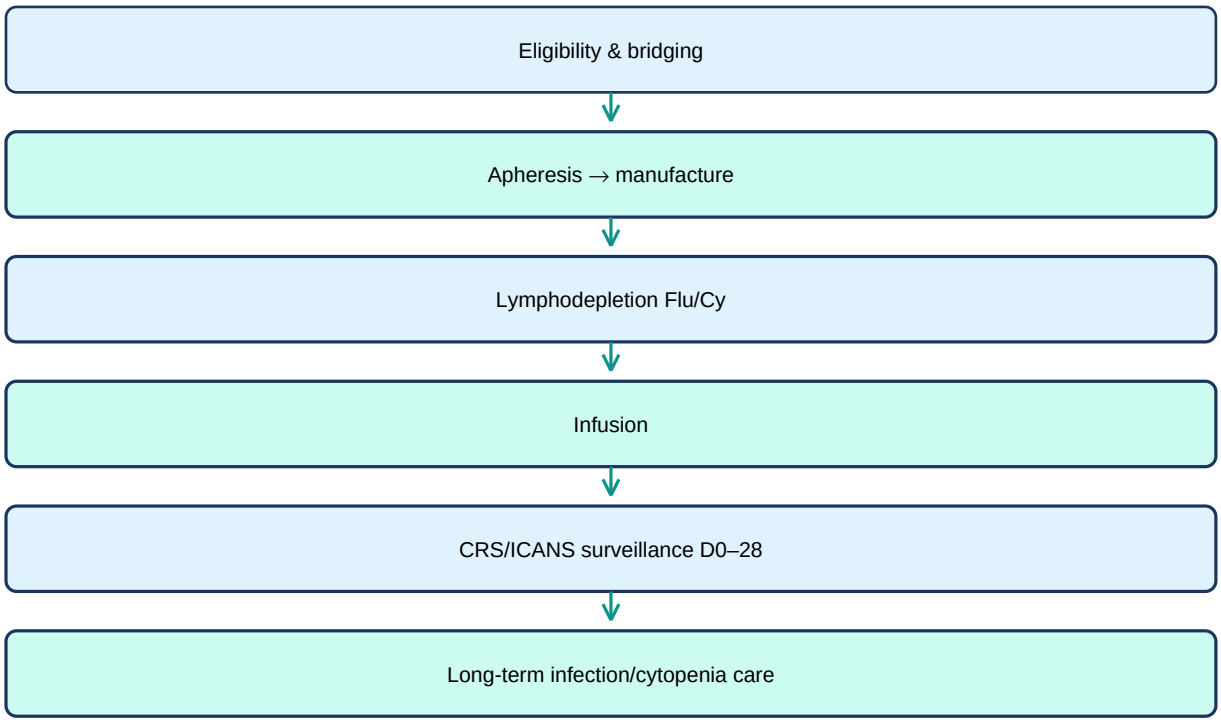

Figure: stepwise exam algorithm

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

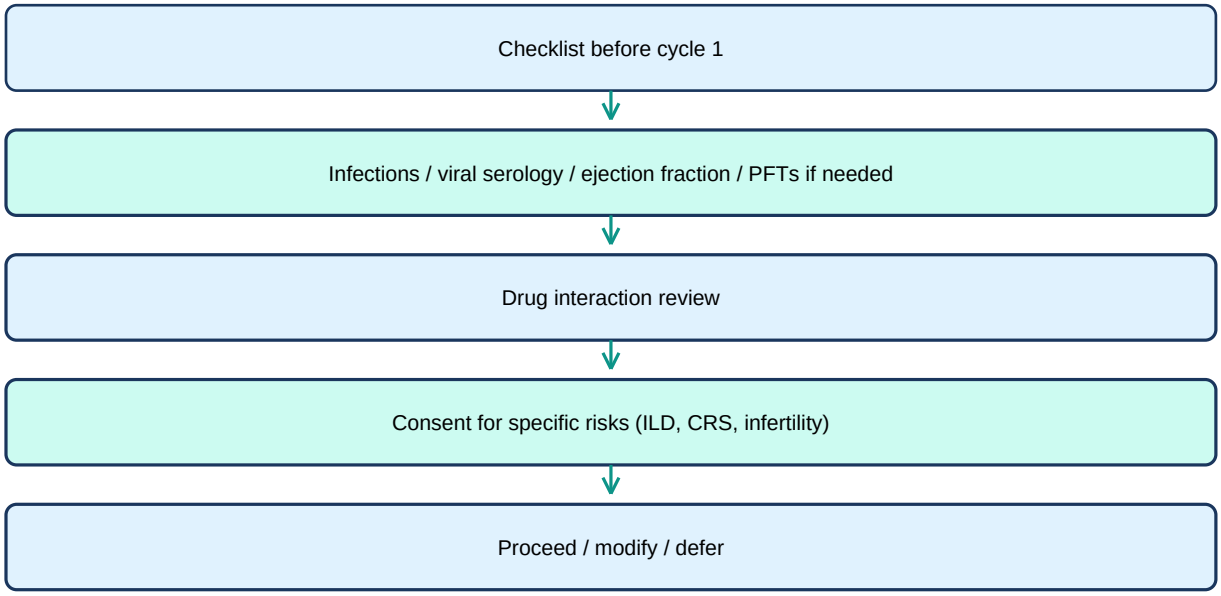

Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.

• Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

• Wall chart + flowchart + trial table = visual completeness.

• Write ‘NCCN 2026’ in quadrant 4 explicitly.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Multiple Myeloma v2.2027; B-Cell Lymphomas v4.2026

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• PD-L1 testing emphasized for gastric/EGJ; limited benefit if PD-L1 negative.

Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.